SPRINT PLANNING & USER STORY WORKSHOP DOCUMENT

Assessment Tool 2 – Agile Sprint Planning & User Story Workshop

1. Project Vision and Stakeholders

1.1 Project Vision

To build a centralized, real-time emergency response platform that connects affected people, rescue agencies, and resource coordinators during natural disasters. The platform prioritizes urgent incidents, speeds up rescue operations, and improves the use of available resources.

1.2 Objectives

Register and locate disaster incidents quickly.

Prioritize emergency requests based on severity and life-threatening risks.

Track rescue teams, medical units, vehicles, and relief supplies.

Monitor shelter locations, capacity, and inventory.

Provide real-time updates and operational reports.

1.3 Stakeholders and Roles

Stakeholder	Role / Responsibility
Affected citizens	Report emergencies and request rescue or shelter.
Emergency dispatch operators	Register and validate incident reports and monitor the dispatch queue.
Rescue teams / field responders	Attend incidents, receive assignments, and update rescue status.
Disaster response coordinators	Coordinate agencies, prioritize incidents, and allocate resources.
Shelter and logistics managers	Manage shelter capacity, food, medicines, and relief supplies.

Police, fire, and medical services	Provide emergency, rescue, and medical support.
System administrator	Manage users, access permissions, and system settings.

2. Epics and User Stories

Epic 1: Incident Reporting and Intelligent Prioritization

US1.1 (5 points): As an affected citizen, I want to submit an SOS report with my GPS location and medical status, so that rescue teams can locate and help me quickly.

US1.2 (8 points): As an emergency dispatch operator, I want the system to prioritize incidents based on life-threatening risks and age-related risk factors, so that the most urgent cases receive help first.

Epic 2: Field Resource and Rescue Team Management

US2.1 (8 points): As a disaster response coordinator, I want to view nearby available rescue teams on an interactive map, so that I can assign a suitable team to high-priority incidents.

US2.2 (3 points): As a field rescue team leader, I want to receive dispatch alerts and update task statuses, so that the control centre has real-time information.

Epic 3: Shelter and Relief Inventory Tracking

US3.1 (5 points): As a shelter manager, I want to update bed availability and food supplies, so that affected people can be directed to shelters with available space.

US3.2 (3 points): As a disaster response coordinator, I want to generate automated response reports, so that agencies can review response times and improve future operations.

3. Acceptance Criteria (Given–When–Then)

US1.1 – SOS Incident Reporting

Given an affected citizen opens the mobile/web interface.

When the citizen submits an SOS report with the situation, number of trapped people, and live location enabled.

Then the system stores the incident, captures GPS coordinates within 10 metres accuracy, creates a unique Incident Ticket ID, and sends the report to the dispatch queue within 3 seconds.

US1.2 – Automated Priority Engine

Given emergency reports are waiting in the queue.

When the system detects high-risk flags such as a medical emergency, rising water, or elderly/children trapped.

Then it marks the incident Priority Level 1 (Critical), moves it to the top of the dispatch list, and displays a red warning.

US2.1 – Interactive Resource Dispatch Map

Given a coordinator is viewing the active operations dashboard.

When the coordinator selects a Priority Level 1 incident on the map.

Then the map highlights available rescue teams within 15 km and displays their equipment and estimated arrival time (ETA).

US2.2 – Responder Status Updates

Given a rescue team has received an assignment.

When the team leader changes the task status to En-route, On-site, or Resolved.

Then the updated status is saved and shown to the control centre without requiring a manual refresh.

US3.1 – Shelter Capacity Tracking

Given a shelter manager is signed in.

When the manager updates bed availability or food supply levels.

Then the system saves the updated values and displays the current shelter capacity and inventory to authorized coordinators.

US3.2 – Response Reporting

Given response incident and task data are available.

When a coordinator requests a report for a sprint or disaster.

Then the system generates a report containing incident counts, response times, task statuses, and resource usage.

4. Product Backlog Prioritization

The MoSCoW method prioritizes work as Must Have, Should Have, Could Have, or Won't Have in the current release.

Priority	Story ID	Backlog Item	Story Points
Must Have	US1.1	SOS reporting with live GPS	5
Must Have	US1.2	Automated incident prioritization	8
Must Have	US2.1	Interactive map and nearest rescue team selection	8
Must Have	US2.2	Responder status updates	3
Should Have	US3.1	Shelter capacity and logistics tracker	5
Should Have	US3.2	Automated response and audit reports	3
Could Have	US4.1	Offline SOS messaging through SMS gateway	8

Won't Have in this release	US5.1	AI drone aerial damage assessment	13

5. Sprint Backlog and Task Breakdown

Sprint 1 duration: 2 weeks. Selected stories: US1.1, US1.2, US2.1, and US2.2. Total: 24 story points.

Story ID	Task ID	Task Description	Assignee	Hours
US1.1	T1.1.1	Create PostgreSQL/PostGIS incident database schema	DB Engineer	6
US1.1	T1.1.2	Develop incident submission API and GPS capture	Backend Developer	8
US1.1	T1.1.3	Design mobile-friendly SOS submission interface	Frontend Developer	6
US1.2	T1.2.1	Develop prioritization logic based on severity rules	Backend Developer	10
US1.2	T1.2.2	Implement real-time priority queue updates	Backend Developer	8
US2.1	T2.1.1	Integrate an interactive map into the control panel	Frontend Developer	8
US2.1	T2.1.2	Find nearest available rescue teams within a radius	DB/Backend Developer	8
US2.2	T2.2.1	Build responder view for task status updates	Frontend Developer	6
US2.2	T2.2.2	Trigger real-time alerts when status changes	Backend Developer	6

6. Effort Estimation Using Story Points

Planning Poker with the Fibonacci sequence (1, 2, 3, 5, 8, 13, 21) is used. Estimates consider complexity, safety risk, and technical dependencies.

User Story	Points	Justification
US2.2 – Status updates	3	Simple status changes with basic authorization.
US1.1 – SOS and GPS	5	Requires location permission and handling weak network signals.
US1.2 – Priority engine	8	Requires multi-factor risk rules and real-time queue updates.
US2.1 – Geospatial map	8	Requires map integration, distance queries, and live markers.

7. Sprint Goal and Expected Deliverables

7.1 Sprint Goal

Establish an end-to-end incident management process that lets citizens submit SOS requests with GPS locations, prioritizes life-threatening incidents automatically, and helps dispatchers locate nearby rescue teams on a live map.

7.2 Expected Deliverables

SOS Incident Reporting Portal with GPS tagging.

Intelligent Priority Queue that moves critical reports to the top.

Dispatcher map showing incident locations and available rescue teams.

Field Responder Task View for assignments and status updates.

Quality assurance tests targeting more than 85% code coverage.

8. Jira Implementation – Screenshot Evidence

After creating the items in Jira, paste the screenshots below the matching captions. Use Windows + Shift + S to capture each screen, then click below the caption and press Ctrl + V in Word. Keep the text readable and crop out unrelated areas.

Figure 1: Jira Project and Scrum Board

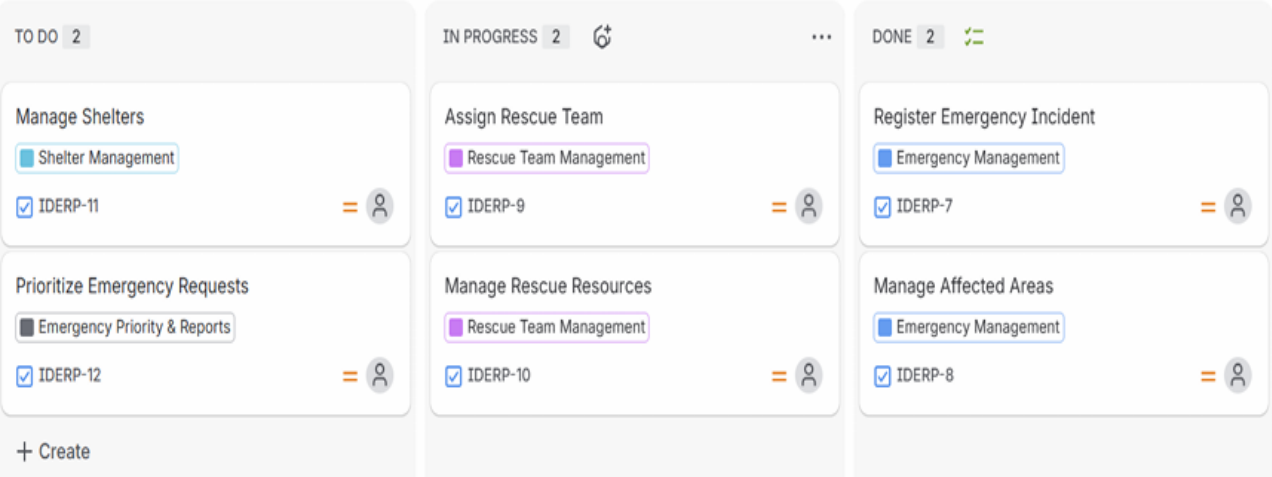


Figure 2: Epics and User Stories

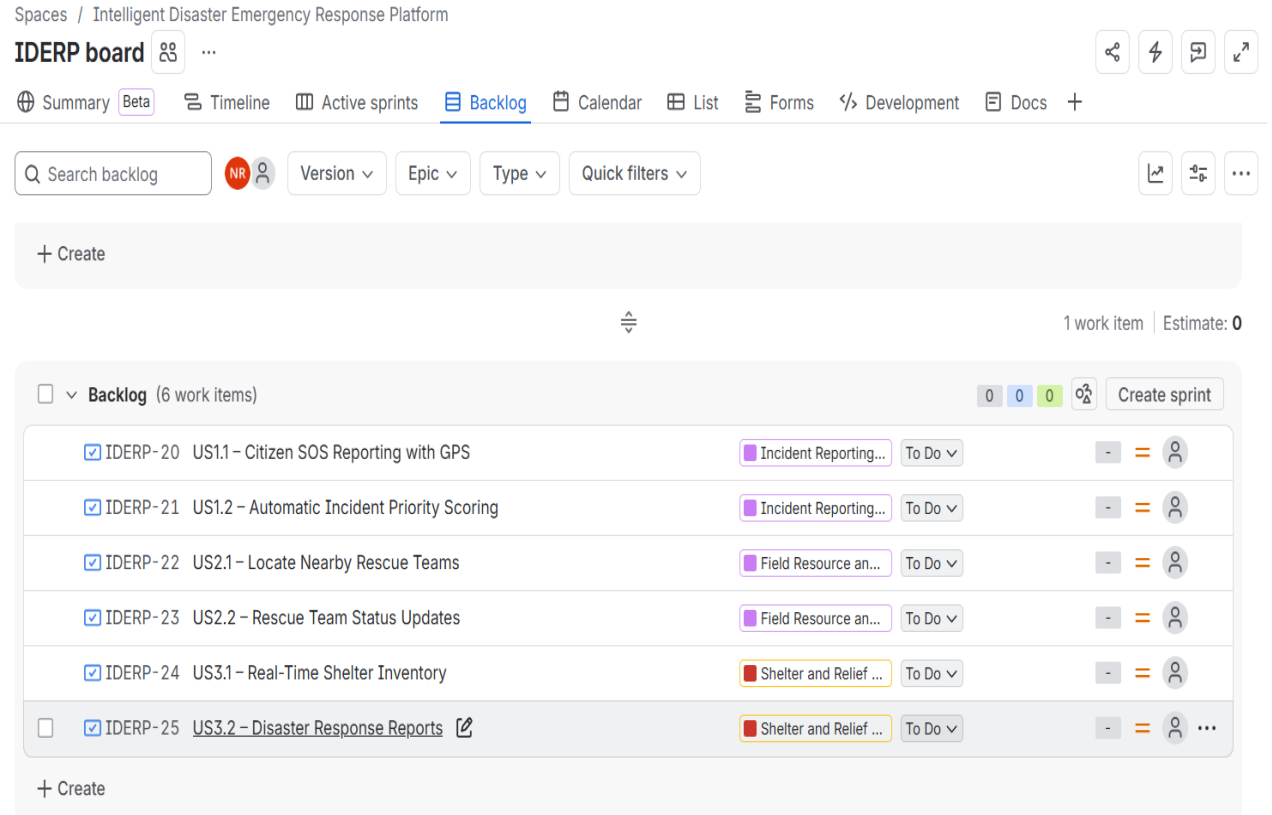


Figure 3: Product Backlog Priorities and Story Points

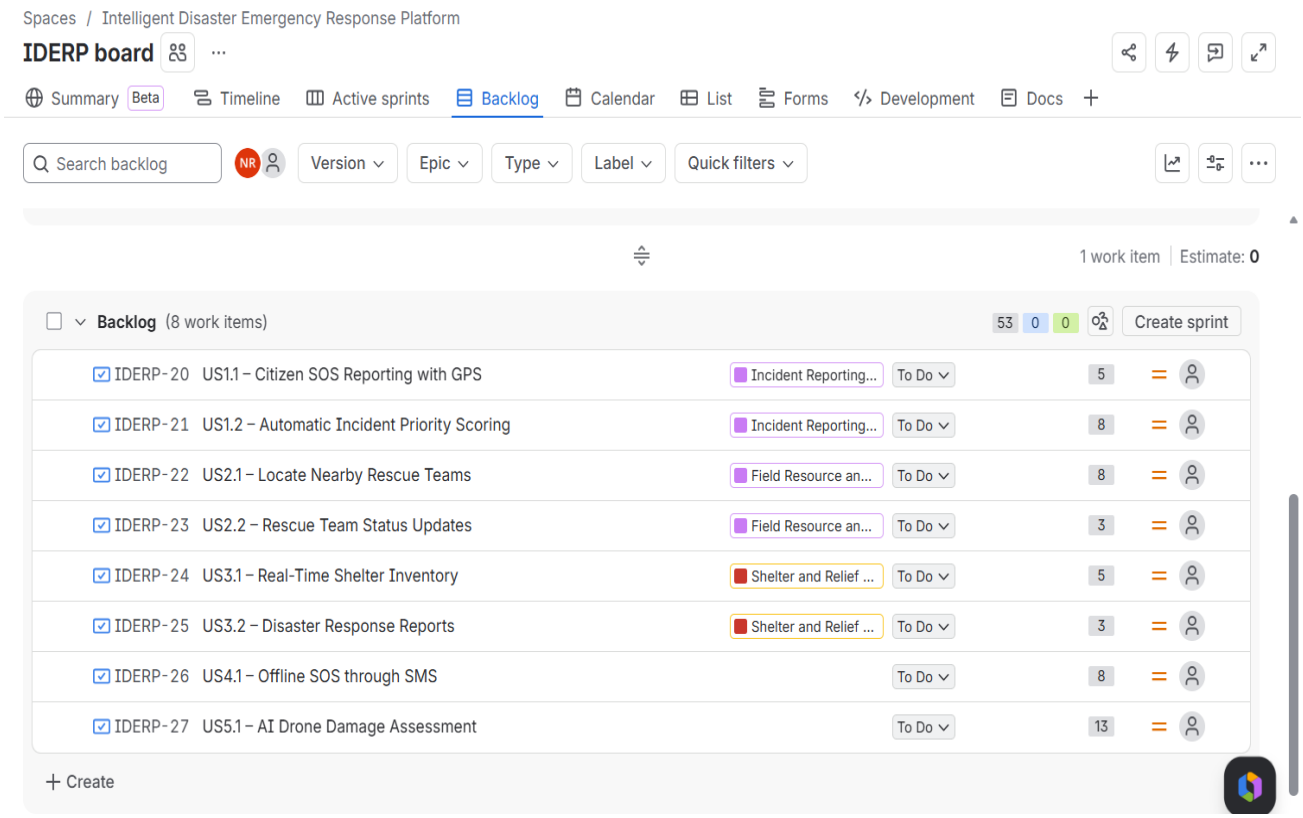


Figure 4: Sprint 1 Planning

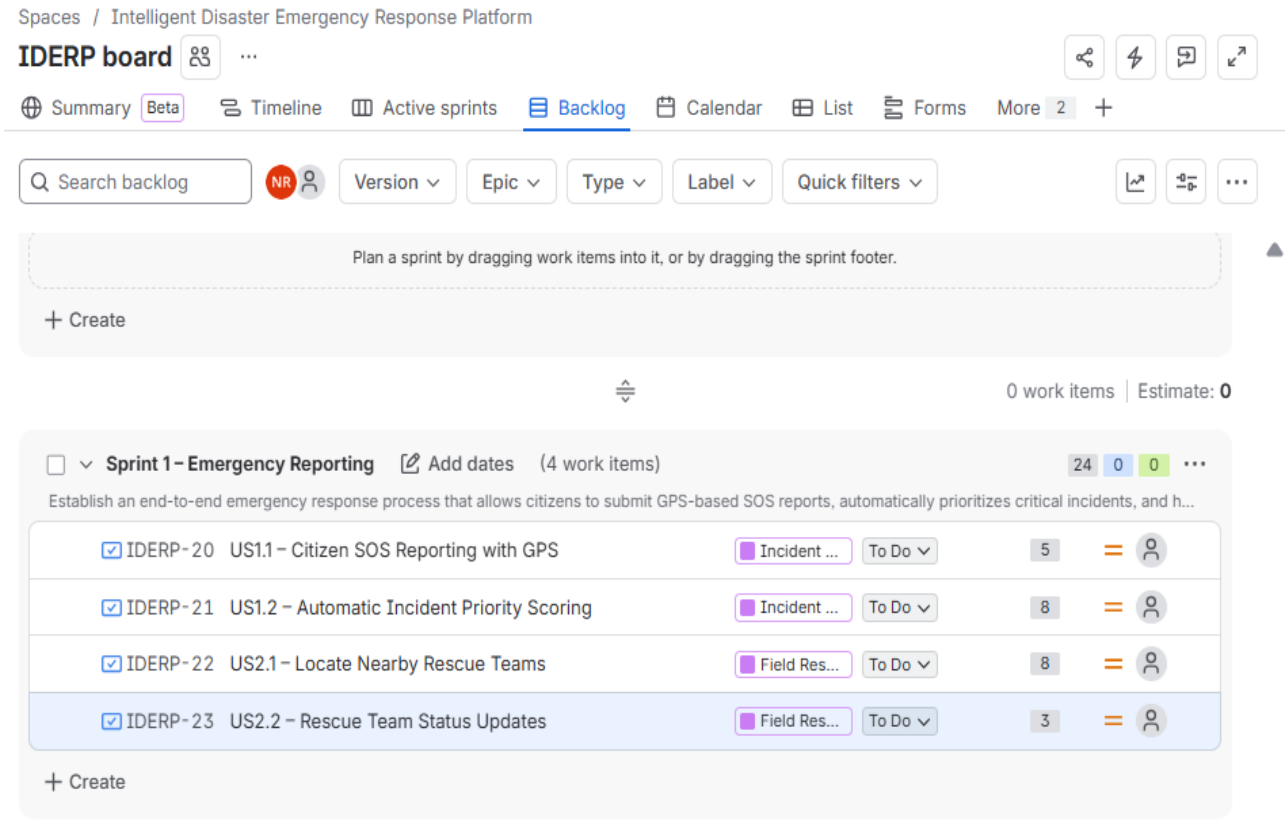


Figure 5: Task Breakdown

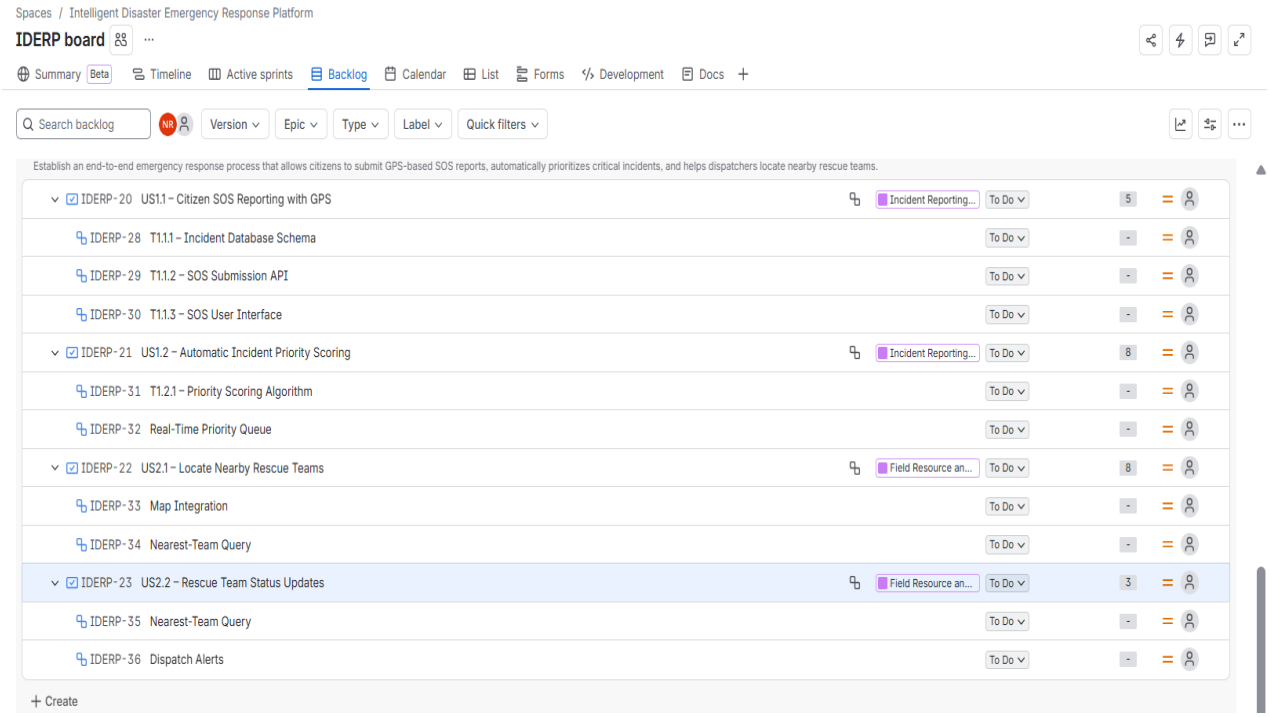
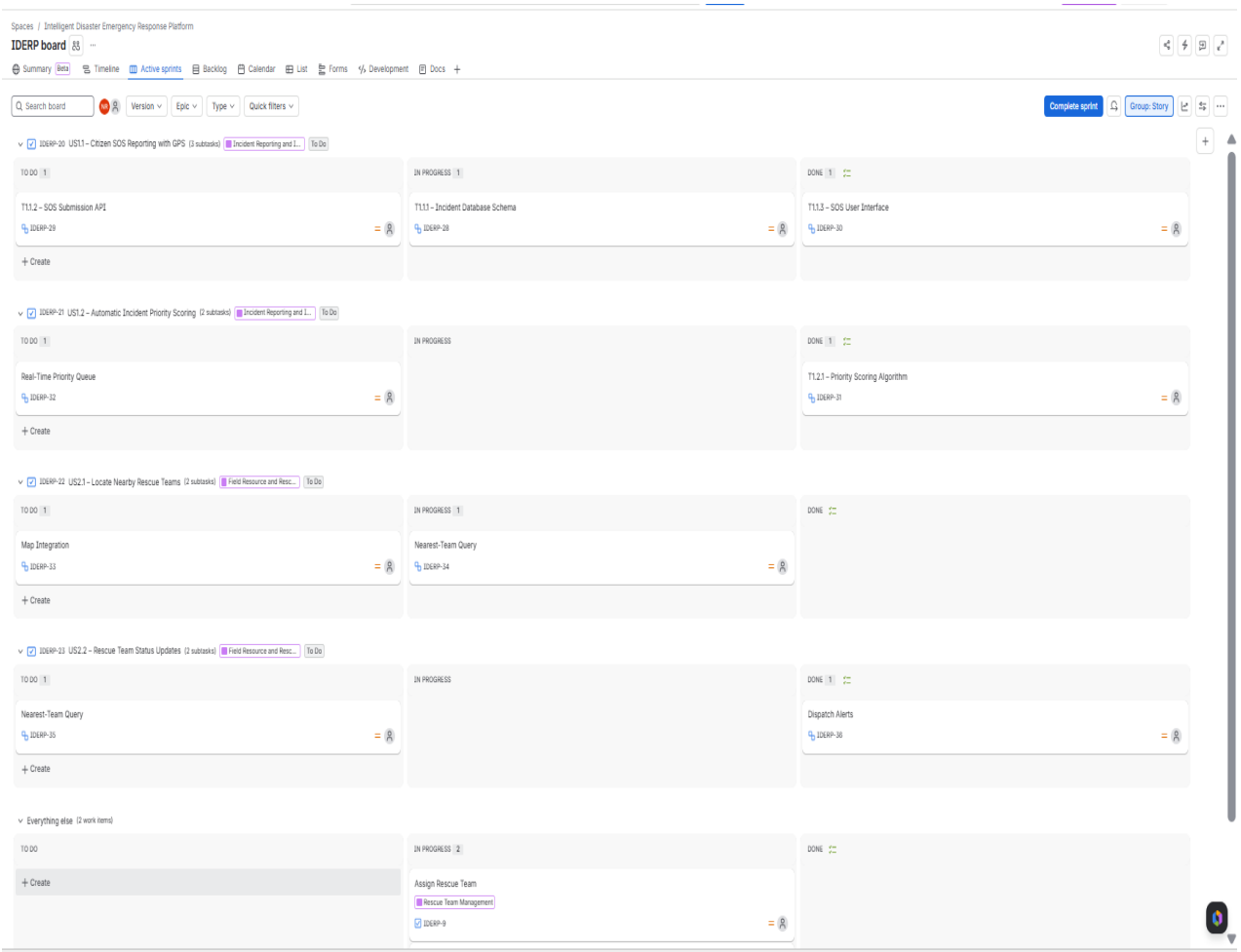


Figure 6: Sprint Board



This sprint planning document organizes the disaster response platform into epics, user stories, acceptance criteria, prioritized backlog items, and development tasks. Planning Poker supports effort estimation, while the two-week Sprint 1 focuses on core incident reporting, prioritization, mapping, and responder updates.